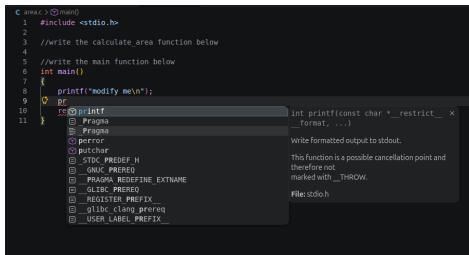


Lab 3: Recursions and Arrays

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VSCode set up (ignore if you have already done this)

VSCode comes with auto-suggestion features:



These are useful for experienced programmers, but are detrimental for learners.

The exam is pencil-and-paper, so it's best to disable all these features.

The next slide explains how.

Open VSCode and press Ctrl+Shift+P (Cmd+Shift+P on Macs).

In the text box, type Preferences: Open User Settings (JSON).

Copy-paste this code (including brackets) into that file:

```
{
"chat.disableAIFeatures": true,
"editor.inlineSuggest.enabled": false,
"editor.quickSuggestions": false,
"editor.suggestOnTriggerCharacters": false,
"editor.wordBasedSuggestions": "off",
"editor.parameterHints.enabled": false,
"editor.snippetSuggestions": "none",
"editor.tabCompletion": "off",
"editor.acceptSuggestionOnEnter": "off",
"editor.acceptSuggestionOnCommitCharacter": false,
"C_Cpp.autocomplete": "disabled"
}
```

Homework 1

Suppose you have £900 deposit in the bank.

The bank pays you interest every year.

The interest is calculated as follows: you get 5% interest on the first £1000, and 2% interest on any amount above £1000.

So each year:

- $\text{new balance} = \text{balance} + \text{interest}$
- if $\text{balance} \leq 1000$, $\text{interest} = \text{balance} * 0.05$
- if $\text{balance} > 1000$, $\text{interest} = 1000 * 0.05 + (\text{balance} - 1000) * 0.02$

How much money do you have after saving for 10 years?

Solve the problem in C using recursion.

Hint: How do you express balance as a function of year?

$$\text{Balance}(0) = 900$$

$$\text{Balance}(1) = \text{Balance}(0) + \text{interest}(\text{Balance}(0))$$

...

$$\text{Balance}(j) = \text{Balance}(j-1) + \text{interest}(\text{Balance}(j-1))$$

Edit the file `deposit.c` to implement this.

Homework 2 (past exam question)

```
#include <stdio.h>

void g(){
    int b = 4, c = 5;
    printf("g is being called!\n");
}

void f(){
    int a = 1, b = 2;
    g();
    printf("g has been called!\n");
}

int main(void){
    f();
    return 0;
}
```

Without running the program, draw the stack memory layout when the program is printing out `g is being called!`.

Verify your answer using the debugger.

Homework 3: summarise some measurements

Given the array

```
double x[] = {2.1, 1.8, 2.4, 2.0, 1.7};
```

modify the `measurements.c` file to write loops in `main` that:

1. calculate and print the sum;
2. calculate and print the mean;
3. count how many observations are greater than 2.0.

Before running, predict each answer.

Homework 4

Suppose you have two matrices $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times l}$.

What is the time complexity of computing matrix multiplication AB ?

1. Express the number of computing cycles needed for this operation using m , n and l .
2. If it takes T seconds to compute AB , how much time would it take to compute CD , $C \in \mathbb{R}^{2m \times 2n}$, $D \in \mathbb{R}^{2n \times 2l}$.

Write your answer and reasoning in the submission text box on blackboard.

Hint: see the next slide for a pseudo-code for matrix multiplication.

To compute $C = AB$, where:

- A is an $(m \times n)$ matrix
- B is an $(n \times l)$ matrix
- C is an $(m \times l)$ matrix

do:

```
for i = 1 to m
  for j = 1 to l
    C[i,j] = 0
    for k = 1 to n
      C[i,j] = C[i,j] + A[i,k] * B[k,j]
```

Submission

Upload your solutions to homeworks 1 and 3 in two separate .c files.

The solution to homework 4 can be written in the submission text box on blackboard.

NOTE Blackboard allows uploading multiple files, so this should work fine.

Before submitting, make sure your code compiles and runs!